

ACD G10-G16 Diaphragm Meters

Compact, high accuracy commercial meter

- ▶ Long term accuracy and reliability
- ▶ Robust, maintenance free meter
- ▶ Compact design
- ▶ High resistant corrosion protection
- ▶ Low Frequency transmitter retrofit

Our new generation Actaris Diaphragm G10-G16 meters combine accuracy and long life in a compact casing. Together with our proven know-how in gas measurement and the use of modern engineering and production techniques we have developed this smaller size, highly accurate meter ensuring reduced shipping costs and easier handling and installation. Our range of ACD G10-G16 meters are designed for commercial use for gas suppliers and gas utilities worldwide.

Principle of Operation

The movement of the diaphragm is caused by the pressure difference between the inlet and the outlet of the meter. The reciprocal filling is controlled by means of two sliding valves. This oscillating movement is transformed into a rotational one and is mechanically transmitted to the totaliser through a magnetic coupling or a stuffing box.

Description

The Actaris Diaphragm G10-G16 meters contain 4 main parts

- 1 A measuring unit with
 - 4 measuring chambers
 - 2 sliding valves
 - and one outlet pipe
- 2 A steel casing fitted with one or two connections
- 3 A magnetic coupling or stuffing box which transmits the movement of the measuring unit to the totaliser
- 4 A totaliser which registers the measured gas

Application

The ACD diaphragm meters are used for applications requiring high precision and large rangeability at low pressure (below 1 bar gauge). They are supplied in two versions – a compact and a standard version (single and two pipe). Due to the volumetric principle of the diaphragm meters, its metrology is not influenced by installation conditions. They are designed for use with natural gas, manufactured gas and other non-corrosive gases. The ACD diaphragm meters are approved for fiscal use.



▶ ACD casing 250 mm



▶ ACD casing 280 mm



▶ ACD measuring unit





► Totaliser standard



► Totaliser with LF "cable"



► Totaliser with LF "binder"



► Thermowell fitted onto an ACD standard

Standard Features

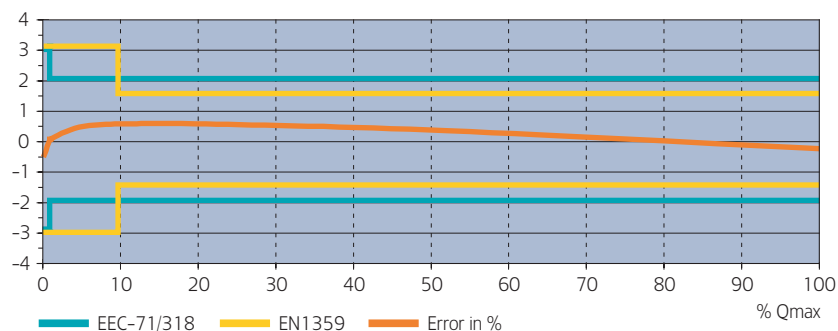
Flow rate range	G10:	from 0.10 m³/h to 16 m³/h
	G16:	from 0.16 m³/h to 25 m³/h
European Metrological Approval (EEC-71/318)	Approval achieved by the German PTB as G6. G10 & G16 (multi-range available) Approval N° PTB-1.42-4016867	
Metrology	In accordance with the EN1359 and MID. Maximum permissible errors are +/-3% from Qmin to 0.1 Qmax and +/-1.5% from 0.1 Qmax to Qmax.	
Starting flow	Typical value:	< 8 dm³/h
Temperature range	Ambient:	-25°C to +55°C
	Gas:	-25°C to +55°C
	Storage temperature:	-40°C to +70°C
Maximum pressure	Compact version:	0.5 bar
	Standard version:	1 bar
Connections	Single pipe or 2 pipe connections From DN32 to DN50 depending on the G-size Other connections are available on request	
Totaliser	8-digit index.	
	UV resistant cover.	
	Fitted with a reflecting disc on the first drum to facilitate periodical checks.	
	Prepared with a magnet in the first or second drum for retrofitting an external Low Frequency transmitter.	
Magnetic coupling	Customised name plate: bar code, customer serial number or logo	
	The meter is equipped as standard with a magnetic coupling.	
Stuffing box	As an alternative a stuffing box can be also installed.	
Reverse flow device	The meter is equipped with a device which prevents registration of reverse flow.	
RPF (Resistant Power Factor)	According to PRS11 (< 1.2)	
Materials	Casing:	steel drawn sheet
	Measuring unit body:	polyacetal (POM)
	Diaphragms:	polyester fabric coated with NBR-ECO
	Distribution valves and grid:	phenol resin
Corrosion protection	Depending on version:	
	- Aluminium-coated + solvent-free paint - Powder coating	
Colour	Light Grey RAL7035	

Options

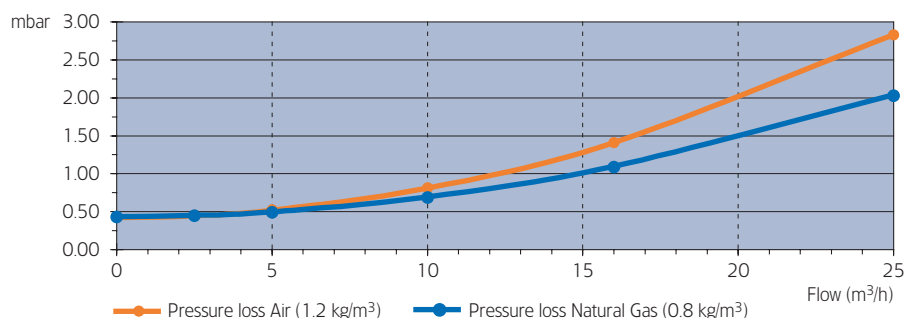
External Low Frequency transmitter	An external Low Frequency transmitter fitted with cable or binder plug can be retrofitted without decommissioning the meters.
Thermowell	The meters can be fitted with a thermowell to allow electronic temperature compensation.
High Temperature Loading (HTL)	The meters can be delivered in a HTL version following EN1359 PNO.1

Metrology

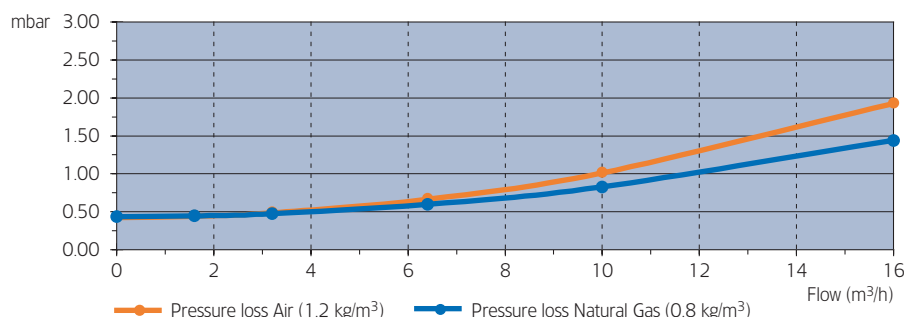
Typical error curve



Pressure loss ACD G16



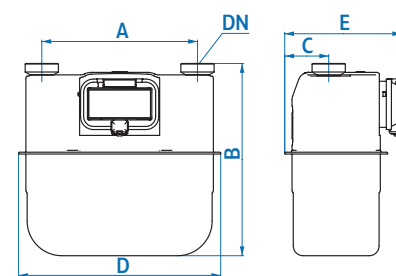
Pressure loss ACD G10



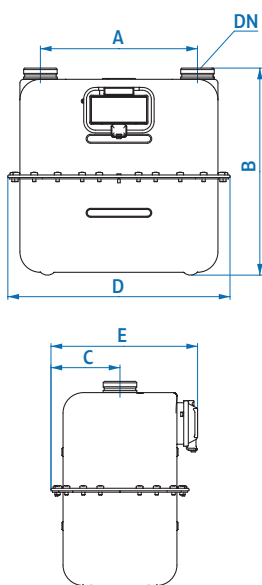
Characteristics

ACD Compact: 2 pipe version

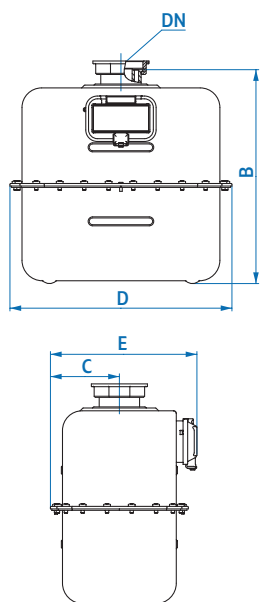
Model	G	Qmax	Qmin	Cyclic volume	DN	Threads	Pmax	Pmax HTL	Pressure loss	A	B	C	D	E	Weight
	Size	m³/h	m³/h	dm³	mm	Standard	bar	bar	mbar	mm	mm	mm	mm	mm	Kg
1	G10	16	0.10	5	32	G1 3/4" ISO228-1	0.5	0.1	1.9	250	309	71	325	189	4
2	G10	16	0.10	5	32	MFIT001	0.5	0.1	1.9	250	309	71	325	189	4
3	G10	16	0.10	5	40	G2" ISO228-1	0.5	0.1	1.9	250	309	71	325	189	4
4	G16	25	0.16	5	40	G2" ISO228-1	0.5	0.1	2.9	250	309	71	325	189	4



► ACD Compact: 2 pipe version



► ACD Standard: 2 pipe version



► ACD Standard: Single pipe version

Characteristics (cont'd)

ACD Standard: 2 pipe version

Model	G Size	Qmax m³/h	Qmin m³/h	Cyclic volume dm³	DN mm	Threads Standard	Pmax bar	Pmax HTL bar	Pressure loss mbar	A mm	B mm	C mm	D mm	E mm	Weight Kg
10	G10	16	0.10	5	32	G1 3/4" ISO228-1	1	0.1	1.9	280	373	123	396	262	7.5
11	G10	16	0.10	5	40	G2" ISO228-1	1	0.1	1.6	280	369	123	396	262	7.5
12	G10	16	0.10	5	40	G2" ISO228-1	1	0.1	1.6	290	394	123	396	262	7.5
13	G10	16	0.10	5	40	G2" ISO228-1	1	0.1	1.6	300	398	123	396	262	7.5
14	G16	25	0.16	5	32	G1 3/4" ISO228-1	1	0.1	2.9	280	373	123	396	262	7.5
15	G16	25	0.16	5	40	G2" ISO228-1	1	0.1	2.7	280	369	123	396	262	7.5
16	G16	25	0.16	5	40	G1 1/2" BS Withworth	1	0.1	2.7	280	373	123	396	262	7.5
17	G16	25	0.16	5	40	G2" ISO228-1	1	0.1	2.7	300	398	123	396	262	7.5
18	G16	25	0.16	5	50	MFIT001	0.2	-	2.7	280	372	123	396	262	7.5
19	G16	25	0.16	5	50	Flange ISO PN10	0.5	-	2.7	280	417	123	396	262	8.2

ACD Standard: Single pipe version

Model	G Size	Qmax m³/h	Qmin m³/h	Cyclic volume dm³	DN mm	Threads Standard	Pmax bar	Pmax HTL bar	Pressure loss mbar	A mm	B mm	C mm	D mm	E mm	Weight Kg
30	G10	16	0.10	5	40	G2 3/4" ISO228-1	1	0.1	1.9	-	382	123	396	262	7.5
31	G16	25	0.16	5	40	G2 3/4" ISO228-1	1	0.1	2.9	-	382	123	396	262	7.5